SOC Analyst AI Agent

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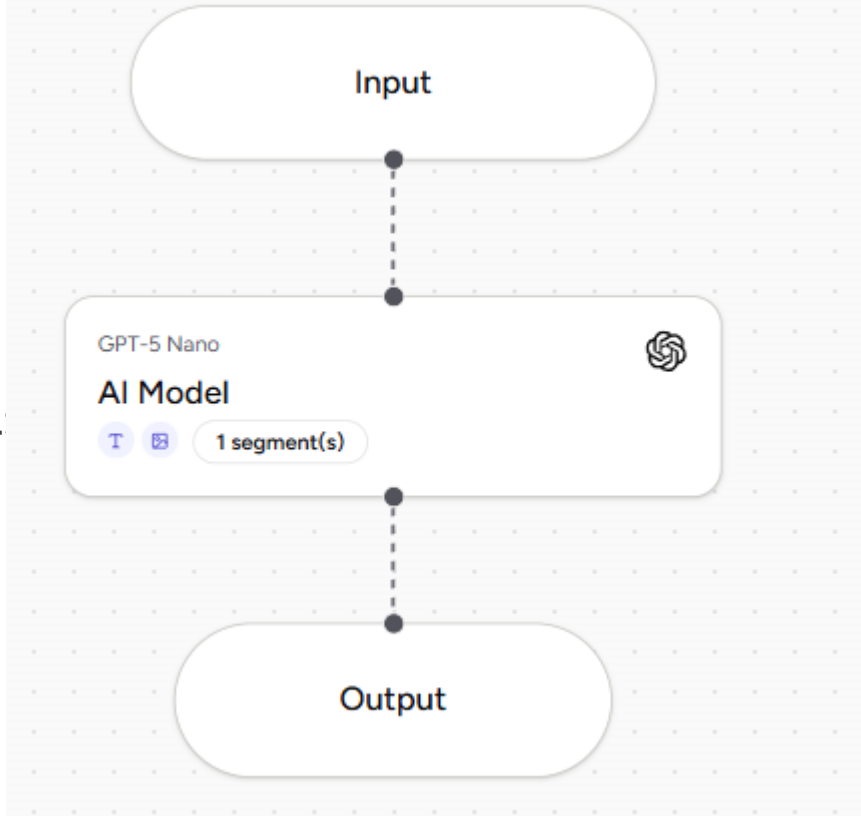
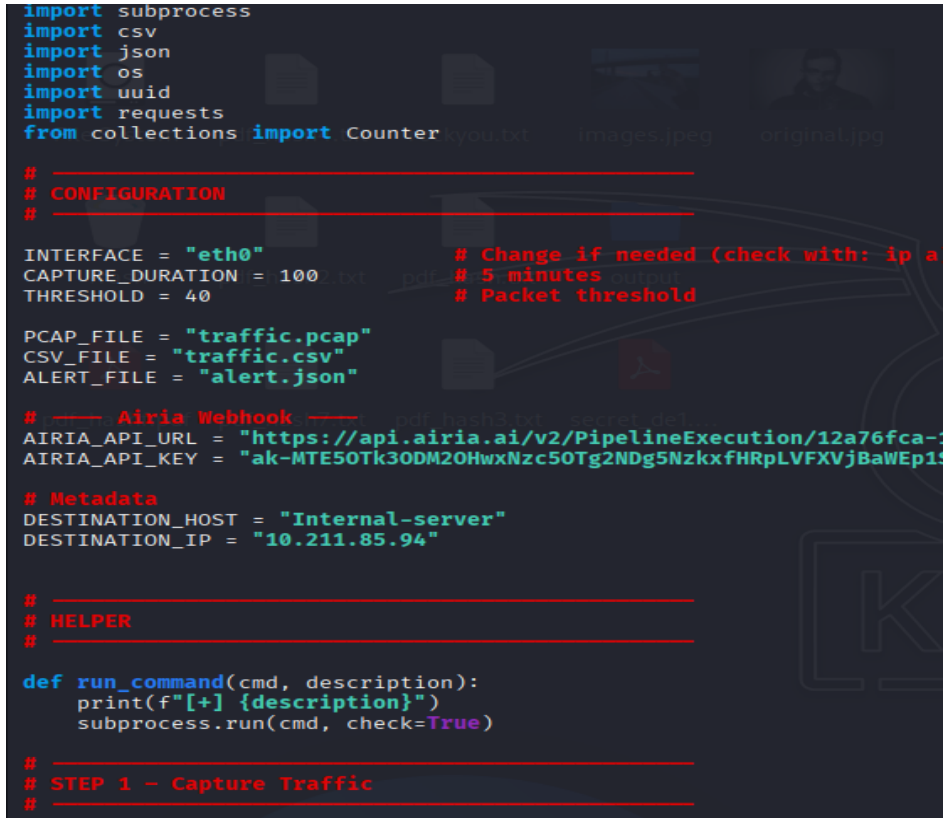
1. Build the AI Agent

- Sign up to airia.ai and create your own project
- Add your own AI model and build four agent
- Publish the agent and get the API detail including the key



2. Run the Python code

- In this step, we will write a Python automation script and run it on the internal server. This script will run a packet capture in the background. The aim is to find malicious IPs and send those in the form of JSON alerts to AIRIA using the API.



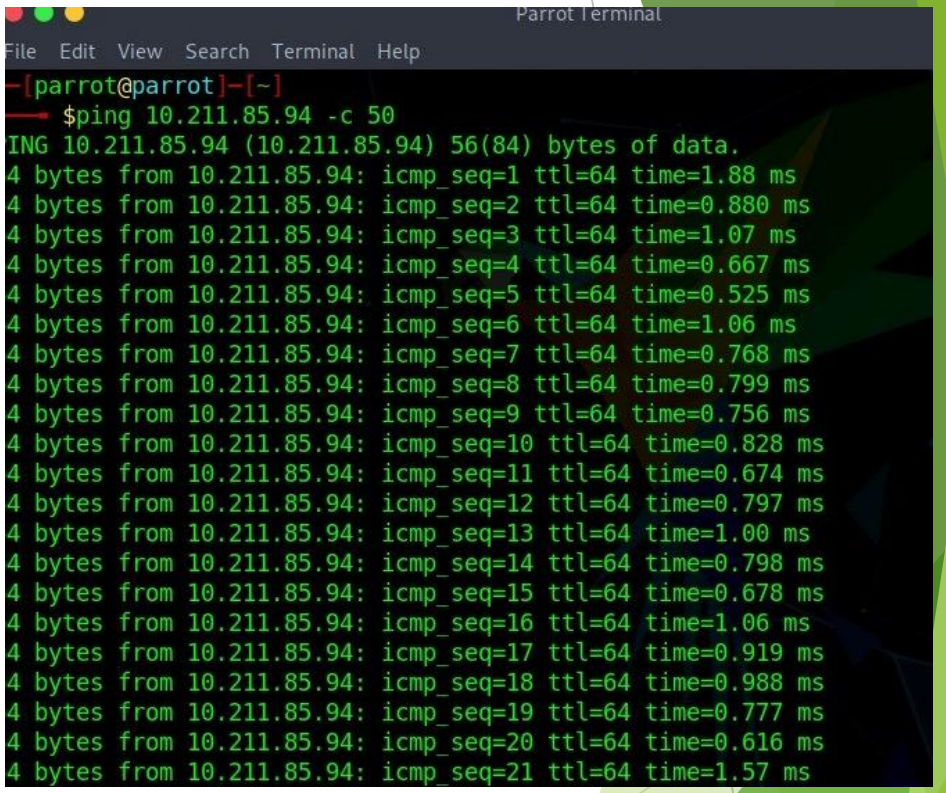
```
import subprocess
import csv
import json
import os
import uuid
import requests
from collections import Counter

# ──────────────────────────────────────────────
# CONFIGURATION
# ──────────────────────────────────────────────

INTERFACE = "eth0"              # Change if needed (check with: ip a
CAPTURE_DURATION = 100          # 5 minutes
THRESHOLD = 40                  # Packet threshold

PCAP_FILE = "traffic.pcap"
CSV_FILE = "traffic.csv"
ALERT_FILE = "alert.json"

# ─── Airia Webhook ───
AIRIA_API_URL = "https://api.airia.ai/v2/PipelineExecution/12a76fca-
AIRIA_API_KEY = "ak-MTE5OTk3ODM2OHwxNzc5OTg2NDg5NzkxfHRpLVFXVjBaWEp1

# Metadata
DESTINATION_HOST = "Internal-server"
DESTINATION_IP = "10.211.85.94"


# ──────────────────────────────────────────────
# HELPER
# ──────────────────────────────────────────────

def run_command(cmd, description):
    print(f"[+] {description}")
    subprocess.run(cmd, check=True)

# ──────────────────────────────────────────────
# STEP 1 — Capture Traffic
# ──────────────────────────────────────────────
```

3. Attack the Server

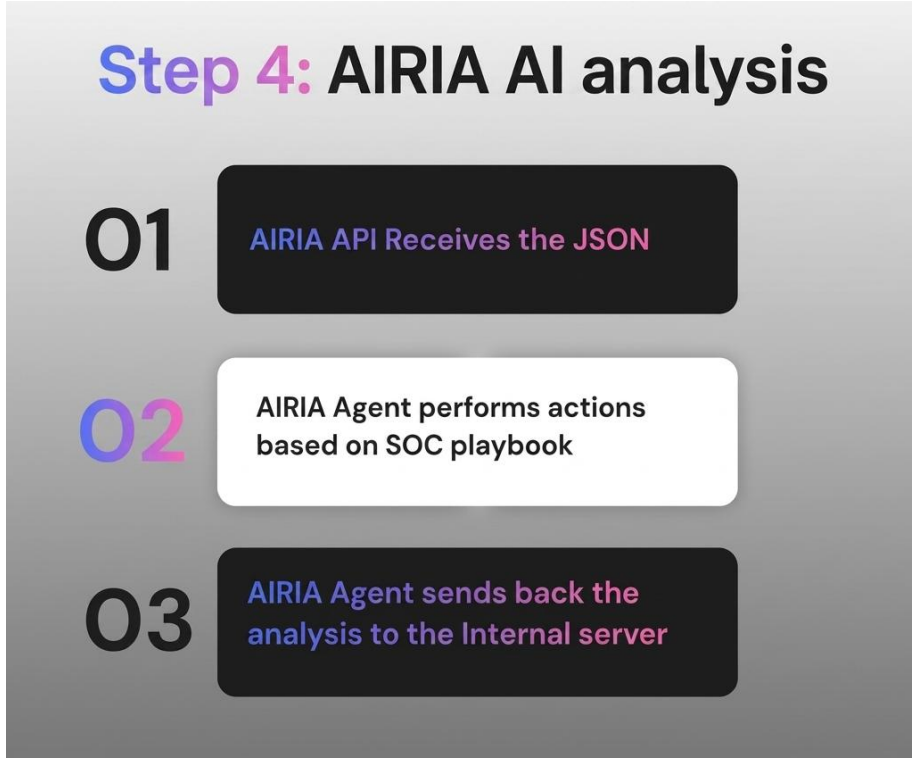
Python Code Flow

- Perform a traffic capture
- Convert it to CSV using tshark
- Find offending IPs based on threshold **> 40**
- Convert data to JSON and send to AIRIA API

From the attacker machine generate some malicious traffic. For this lab, I have used ping only for demonstration purposes. You can try different types of DOS attacks. While the traffic is generated, the packet count will increment on the Internal server as per the logic built into the code.



4. Airia AI analysis



5. Result

It detected the attacker IP as malicious